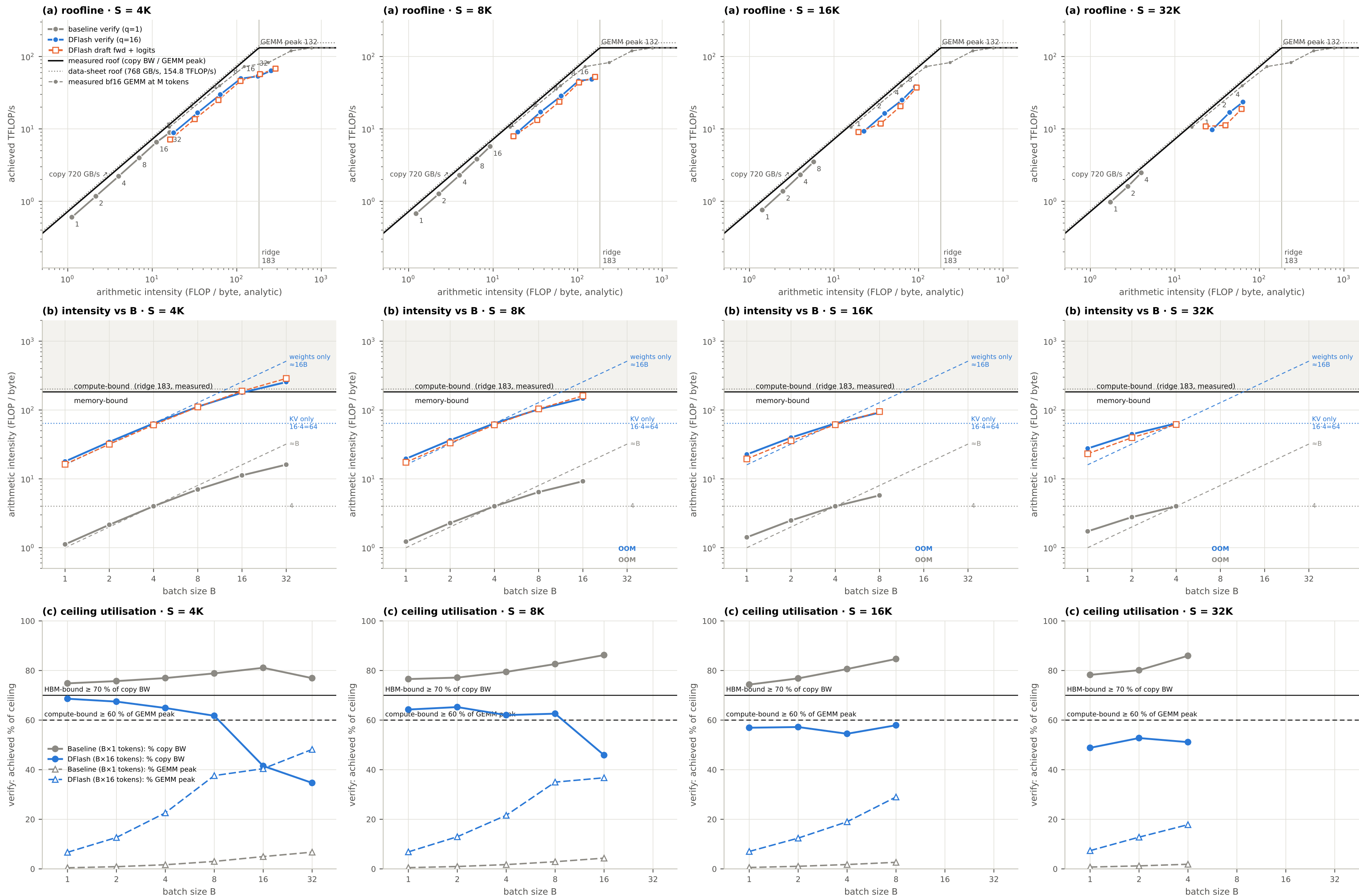


Qwen3-8B — verify forward: memory-bound → compute-bound as the batch grows (RTX A6000, N = 256, EOS suppressed)



Bytes and FLOPs are analytic (plan §5.4): executed target weights once (no embedding table), KV at the mean context $S + N/2$ once per row, GDN state in and out, logits written and read; FLOPs = 2-params-tokens + attention 4-q-L-H_q-d per layer + the delta rule. The draft adds fc and K/V projection of the kept context columns and 15 LM-head rows. Time is the CUDA-event mean per step, over all steps of the run (a finished row still computes). 8B at 16K-32K: KV dominates the bytes, so intensity stays near the KV-only line (64) and the q=16 attention kernel, not HBM, sets the time (see batch_verify_breakdown).